



Adyant Agrawal

Postdoctoral researcher,
Institute for Computational
Physics, Stuttgart

Interests

- Fluid-Structure Interactions
- Interfacial effects
- Hydrodynamics
- Low-Reynolds number flows
- Porous Media Flow
- Multiscale Modelling
- Confined Electrolytes

Methods

- Molecular Dynamics
- Finite Element Methods
- Mesoscale hydrodynamics
- Free-energy methods
- Monte Carlo sampling
- Stochastic Dynamics
- Machine Learning
- Software Development

Tools

- GROMACS
- LAMMPS
- COMSOL
- Python
- MATLAB
- C++
- Git, Github, Gitlab

Education

Ph.D. in Mechanical Engineering

Dec 2018 - Sep 2022

School of Engineering and Material Science
Queen Mary University of London

Thesis title: Liquid exfoliation and rheology of 2D nanomaterials

Thesis supervisor: Prof. Dr. Lorenzo Botto

Bachelor of Technology in Engineering Physics

Jul 2013 - Nov 2017

Department of Physics
Indian Institute of Technology Delhi

Thesis title: Collective motion of self-propelled rods in quasi-2D geometry

Thesis supervisor: Prof. Dr. Sujin Babu

Research Experience

Postdoctoral Researcher

Jan 2023 - Present

Institute for Computational Physics, University of Stuttgart

Developed and applied atomistic, machine-learned, and multiscale simulation methods to investigate confined and interfacial fluid phenomena:

- Salt precipitation and crystal growth analysed using molecular dynamics, with a focus on interfacial hydration and electrostatics.
- CO₂ oversolubility under nanoscale confinement quantified using Wang Landau sampling and continuum modelling.
- Development of a machine-learned implicit solvent (MLIS) model for interfacial and confined electrolyte systems.
- Pressure generation in water confined between salt crystals and silica surfaces quantified using thermodynamic free-energy methods.
- PFAS adsorption at air-solution interfaces informed by free energy calculations and interfacial structuring effects.
- Contributor to MAICOS, an open source analysis framework for structural and electrostatic characterisation of confined molecular systems.

Doctoral Student

Dec 2018 - Sep 2022

Queen Mary University of London

Developed continuum hydrodynamics and atomistic frameworks to investigate fluid-structure interactions in confined and sheared flows:

- Quantitative comparison of continuum theory and atomistic simulations of viscous peeling dynamics of graphene sheets immersed in liquids.
- Flow-induced orientational dynamics and deformation of 2D nanomaterials in shear flow, elucidating elasto-hydrodynamic coupling and suspension rheology.
- Modelling rheology of semi-dilute graphene suspensions, linking nanoscale slip and inter-particle interactions to macroscopic viscosity.

Bachelor's Student

May 2016 - Nov 2017

Indian Institute of Technology Delhi

Developed computational models to investigate collective dynamics, hydrodynamic interactions, and transport phenomena in active and cosmological systems:

- Mesoscale modelling of collective motility of micro-swimmers using Taylor-line representations coupled with Multi-Particle Collision Dynamics (MPC-AT+a).
- Simulation of collective dynamics of self-propelled polar rods in quasi-2D geometries, investigating trapping induced by geometric constraints.
- Cosmological N-body simulations of structure formation in the intergalactic medium, modelling Ly- α absorption spectra.

Certifications

2018
Advanced Training in Statistical Physics
Freie Universität Berlin

2022
Diploma of Researcher Development
Queen Mary University of London

Positions of Responsibility

2020 - 2021
Program Leader
Eklavya Underprivileged Education Trust, India

2015 - 2016
Department Convener
Co-curricular and Academic Interaction Council, IIT Delhi

2015 - 2016
Batch Representative
Board for Academic Programs, IIT Delhi

2014 - 2015
Student Representative
Board for Student Welfare, IIT Delhi

Awards & Academic Distinctions

- 2025 **Best Poster**, Condensed Matter and Quantum Materials Conference, Bristol
Awarded for: "Negative intrinsic viscosity in graphene suspensions: insights from molecular dynamics and continuum theory."
- 2022 **Queen Mary Doctoral Research Studentship** in the School of Engineering and Materials Science
- 2017 **Summer Research Fellowship**, Indian Institute of Science Education and Research (IISER) Mohali, India
- 2015 **Formula Student Design Challenge**, Electronics Engineer — Team rank 9
- 2013 **All India Percentile 99.5**, IIT Joint Entrance Examination

Supervision and Teaching

- 2024–2026 **Co-supervisor** – PhD student Russell Kajouri (CO₂ oversolubility under nanoscale confinement)
- 2024–2025 **Tutor and Course Developer** – Advanced Simulation Methods II, University of Stuttgart
- 2023–2024 **Course Instructor** – Advanced Simulation Laboratory (Biophysical Molecular Dynamics), University of Stuttgart
- 2021–2022 **Laboratory Demonstrator** – Solid Mechanics Laboratory, Queen Mary University of London
- 2020–2022 **Teaching Assistant** – Solid Mechanics, Queen Mary University of London

Publications

- **A. Agrawal**, C. Kamal, S. Gravelle, and L. Botto, Negative intrinsic viscosity in graphene nanoparticle suspensions induced by hydrodynamic slip, *ACS Nano* **19**, 38350–38359 (2025). DOI: 10.1021/acsnano.5c10415.
- **A. Agrawal**, S. Gravelle, C. Holm, and A. Schlaich, Interfacial ion hydration and electrostatics govern salt precipitation and crystal morphology, *J. Chem. Phys.* **163** (2025). DOI: 10.1063/5.0299741.
- **A. Agrawal**, S. Gravelle, C. Kamal, and L. Botto, Viscous peeling of a nanosheet, *Soft Matter* **18**, 3967–3980 (2022). DOI: 10.1039/D1SM01743H.
- **A. Agrawal** and S. B. Babu, Self-organization in a bimotility mixture of model microswimmers, *Phys. Rev. E* **97**, 020401(R) (2018). DOI: 10.1103/PhysRevE.97.020401.
- **A. Agrawal**, C. Holm, and A. Schlaich, Interfacial stress generation during crystal growth in silica nanoconfinement, (*in preparation*).
- H. Stooß, P. Loche, **A. Agrawal**, M. Becker, K. Fischer, S. Gravelle, P. Stärk, and A. Schlaich, MAICoS: A toolkit for the Molecular Analysis of Interfacial and Confined Systems (*in preparation*).
- W. Zhang, **A. Agrawal**, S. Kondrat, A. Schlaich, C. Holm, and B. Guo, A predictive model of PFAS adsorption under complex water chemistry (*in preparation*).
- R. Kajouri, **A. Agrawal**, P. Stärk, A. Schlaich, and C. Holm, An extended Langmuir adsorption model for predicting gas uptake in liquid-filled slit pores (*in preparation*).

Contributed Talks

- 2025 **The role of hydration and electrostatics in salt crystal growth** – 9th International Soft Matter Conference, Minoa Palace, Crete, Greece
- 2025 **Salt precipitation in porous media: atomistic coupling of hydration and electrostatics** – CECAM International Symposium 2025: Molecular Modelling in Soil Science – Achievements and Perspectives
- 2025 **Molecular insights into irregular growth of salt crystals: the role of charge and water structure** – Condensed Matter and Quantum Materials (CMQM) Conference 2025, University of Bristol, UK
- 2024 **The interplay between hydration and growth of a salt crystal** – Liquid Matter Conference 2024, Johannes Gutenberg University Mainz, Germany
- 2024 **Role of hydration and electrostatics in salt precipitation** – DPG Spring Meeting 2024, Berlin, Germany
- 2022 **Viscous peeling of graphene sheets: towards producing graphene on the ton scale** – European Solid Mechanics Conference 2022
- 2021 **Flow-induced dispersion of 2D nanomaterials in sheared liquids** – 4th Russell Binions Research PhD Symposium, Queen Mary University of London
- 2021 **Viscous peeling of a graphene sheet** – APS March Meeting 2021
- 2021 **Can simulations help improve graphene production?** – Industrial Engagement Event 2021, Queen Mary University of London
- 2018 **Swimmers with different motility self-segregate** – Computational Principles to Organise Complexity, Summer School, Nice, France

Contact

- 📍 70174 - Stuttgart
- ☎ +49 015141221031
- ✉ adyant2@gmail.com
- ✉ aagrawal@icp.uni-stuttgart.de